the portion OA of the curve is also reversible. As the field increases, a greater number of domains grow favourably which results in a large increase in magnetization (portion AB). The boundary displacements in this region are often large and irreversible. The growth of domains continues until the favourable domains grow up to the maximum extent with their magnetization vectors still pointing along the so-called *easy directions* of magnetization. As the field increases further, the domains rotate from their easy directions to the direction of the applied field; the magnetization increases slowly (portion BC) and finally attains a saturation value $M_s$ when all the domains point along the direction of the field. On decreasing the field, the magnetization does not follow the same path because the aligned domains do not regain their random state of orientation easily. There exists some non-zero magnetization even after removing the field altogether. This magnetization is called the *remanent magnetization* or *remanence,* $M_r$. The magnetization can be reduced to zero by applying a reverse magnetic field known as the *coercive field* or *coercivity,* $H_c$. A similar variation in the reverse magnetization is observed as the reverse field is first increased and then decreased. The closed loop CEFKC is called the *hysteresis loop.* A similar loop is obtained by plotting $B$ versus $H$ except that the line CI in this curve is never parallel to the field axis. The hysteresis curves are quite important in determining the quality of a magnetic material and selecting the material for a particular application.

## 8.6 ANTIFERROMAGNETISM

Antiferromagnetism originates when the spin moments of the neighbouring atoms are ordered in an antiparallel arrangement as shown in Fig. 8.8 or when the exchange integral is negative. A crystal exhibiting antiferromagnetism may be considered to be consisting of two interpenetrating sublattices A and B, one of which is spontaneously magnetized in one direction and the other is spontaneously magnetized in the opposite direction. This type of magnetism was first observed in the crystals of MnO. In the absence of an external magnetic field, the neighbouring magnetic moments cancel out each other and the material as a whole exhibits no magnetization. However, when a field is applied, a small magnetization appears in the direction of the field which increases further with temperature. Such a behaviour is typical of an antiferromagnetic material. The magnetization becomes the maximum at a critical temperature $T_N$, called the *Neel temperature,* which is analogous to the Curie temperature in the paramagnetic or ferromagnetic substances. Above this temperature, the magnetization decreases continuously which is indicative of the paramagnetic state of the material. The variation of susceptibility with temperature is shown in Fig. 8.9 and is compared with the corresponding variations for the paramagnetic and ferromagnetic substances. Unlike the ferromagnetic case, the susceptibility

of an antiferromagnetic substance is not infinite for $T$ equal to $T_N$, but has a weak cusp. The antiferromagnetic behaviour can be explained with the help of the molecular field theory which was initially developed by Neel. A brief description of this theory is given below.

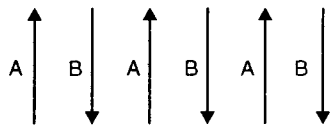



Fig. 8.8. *Neighbouring antiparallel spin moments of A and B sublattices producing antiferromagnetism.*

Consider an antiferromagnetic substance consisting of two sublattices A and B such that an atom of one type has all the nearest neighbours of the opposite type. The internal or *molecular field* acting on an atom can be obtained by using the molecular field approximation developed by Weiss in which the interaction between the various magnetic moments is considered. In an antiferromagnetic substance, besides the nearest neighbour negative AB interaction, there also exists the next nearest neighbour negative AA (or AB) interaction which is quite weak as compared to the former. If $\alpha$ and $\beta$ represent the interaction parameters, i.e., the Weiss field constants for AA (or BB) and AB interactions respectively, the internal fields present at A and B sites, expressed in terms of flux densities, can be written as

$$\left.\begin{aligned} B_{mA} &= -\alpha M_A - \beta M_B \\ B_{mB} &= -\beta M_A - \alpha M_B \end{aligned}\right] \qquad (8.76)$$

where $M_A$ and $M_B$ are the magnetizations of the corresponding sites. In the presence of an external field $B$, the effective fields at $A$ and $B$ sites become

$$\left.\begin{aligned} B_A &= B - \alpha M_A - \beta M_B \\ B_B &= B - \beta M_A - \alpha M_B \end{aligned}\right] \qquad (8.77)$$

To determine the total magnetization, $M_A+M_B$, of the specimen, we consider the temperature regions below and above the Neel temperature.

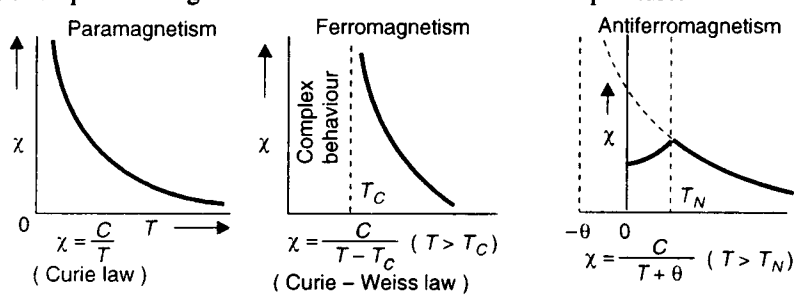



Fig. 8.9.

(*i*) $T>T_N$

In this region, the specimen behaves as paramagnetic; the magnetizations of the two sublattices can be written in accordance with Eq. (8.33) as

$$\left.\begin{aligned} M_A &= \frac{N\mu^2}{3kT} B_A \\ M_B &= \frac{N\mu^2}{3kT} B_B \end{aligned}\right] \qquad (8.78)$$

where

$$\mu^2 = \mu_B^2 g^2 J(J+1).$$

In these expressions, it has been assumed that the $A$-type atoms are identical with $B$-type atoms and their densities are equal, i.e., $N_A = N_B = N$. The total magnetization $M$ is, therefore, obtained as

$$\begin{aligned} M &= M_A + M_B \\ &= \frac{N\mu^2}{3kT}(B_A + B_B) \\ &= \frac{N\mu^2}{3kT}\left[2B - (\alpha+\beta)(M_A + M_B)\right] \qquad \text{[Using Eq. (8.77)]} \\ &= \frac{N\mu^2}{3kT}\left[2B - (\alpha+\beta)M\right] \qquad (8.79) \end{aligned}$$

This yields the expression for susceptibility as

$$\chi = \frac{\mu_0 M}{B} = \frac{\dfrac{2\mu_0 N\mu^2}{3k}}{T + \dfrac{N\mu^2(\alpha+\beta)}{3k}} = \frac{C}{T+\theta} \qquad (8.80)$$

where

$$C = \frac{2\mu_0 N\mu^2}{3k}$$

and

$$\theta = \frac{N\mu^2(\alpha+\beta)}{3k} \qquad (8.81)$$

The expression (8.80) satisfies the experimental data on susceptibility quite well. It is analogous to the Curie-Weiss law given by Eq (8.64) for ferromagnetic materials above the critical temperature except that it contains the term $T + \theta$ instead of $T-\theta$ (or $T-T_C$) in the denominator. The $1/\chi$ versus $T$ curves are shown in Fig. 8.10 to compare the antiferromagnetic behaviour with the

paramagnetic and ferromagnetic ones. It can be shown that the Neel temperature, $T_N$, is not identical with $\theta$ and the two are related as

$$\frac{T_N}{\theta} = \frac{\beta - \alpha}{\beta + \alpha} \qquad (8.82)$$

Thus the value of $T_N$ depends on the strength of *AB* and *AA* (or *BB*) interactions. For $\alpha = 0$, $T_N = \theta$. If $\alpha$ is positive, $T_N < \theta$. Also, $T_N$ increases with $\beta$. These results have been verified experimentally.

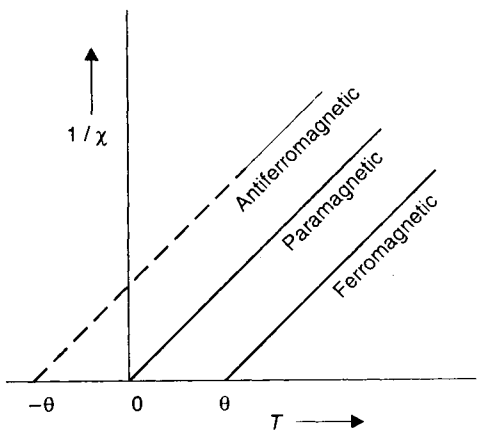



Fig. 8.10. *Plot of $1/\chi$ versus T for para-, ferro- and antiferromagnetic substances above their critical temperatures.*

(ii) $T < T_N$

In this temperature region, each of the two sublattices *A* and *B* is spontaneously magnetized along its preferred direction of magnetization, but the net magnetization is still zero. The application of an external field, however, results in some net megnetization. There are two directions of the applied field which are of particular interest — one perpendicular and the other parallel to the preferred direction. The susceptibility in these two directions are determined separately and the average is taken to calculate the net susceptibility It can be shown that the susceptibility in the perpendicular direction, $\chi_\perp$, is independent of temperature, i.e., remains almost constant

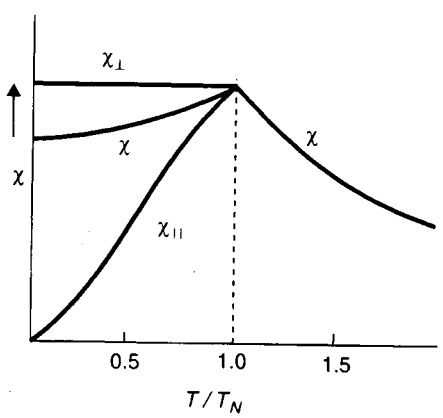



*Fig. 8.11 Plot of $\chi$ versus $T/T_N$ for an antiferromagnetic single crystal. For $T<T_N$, $\chi$ is also a function of direction of external field relative to the preferred direction of magnetization.*

below the Neel temperature. The susceptibility in the parallel direction, $\chi_{\parallel}$, decreases continuously with decrease in temperature and approaches zero as $T \rightarrow 0$ K. This means that at 0 K, the internal field is so strong that it does not allow the spins of one sublattice to change their orientation along the spins of the other sublattice even when an external field is present. At $T=T_N$, $\chi_{\perp} = \chi_{\parallel} = \chi$. The variations of $\chi_{\perp}$, $\chi_{\parallel}$ and $\chi$ with temperature are shown in Fig. 8.11.

## 8.7 FERRIMAGNETISM

Ferrimagnetism is identical to antiferromagnetism except that the magnetization of the two sublattices have different magnitudes which result in a non-zero value of net magnetization. This type of magnetism occurs in materials such as ferrites which are basically the oxides of various metal elements. The most common example is the one of magnetite or ferrous ferrite, $Fe_3O_4$ or $FeO.Fe_2O_3$. It has been found that for this particular ferrite, one $Fe^{3+}$ ion occupies tetrahedral or A-site, i.e., it is tetrahedrally coordinated to four oxygen anions while the other $Fe^{3+}$ ion and the $Fe^{2+}$ ion occupy octahedral or B-sites, each being octahedrally coordinated to six oxygen anions. The ions present at A and B sites constitute A and B sublattices respectively and have opposite types of magnetizations. The net magnetization of a formula unit of $Fe_3O_4$ is, therefore, equal to the magnetization of a single $Fe^{2+}$ ion and is represented in Fig. 8.12. However, one unit cell of magnetite having the spinel structure contains eight formula units of $Fe_3O_4$ in it and accordingly the net magnetization of a unit cell is equal to the sum of the magnetizations of eight $Fe^{2+}$ ions.

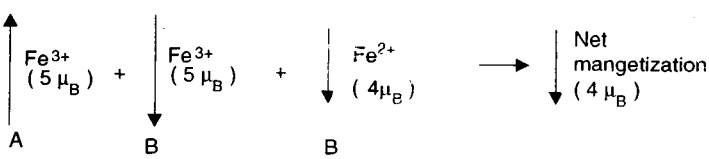



Fig. 8.12. *Magnetizations of the individual sites and the net magnetization in one formula unit of $Fe_3O_4$.*

It is thus easy to note that a ferrimagnetic solid resembles a ferromagnetic one in the sense that both possess spontaneous magnetization. Also, both exhibit hysteresis and almost identical magnetic properties. Neel attributed the antiparallel arrangement of spins of A and B sites to the negative AB interaction. Besides this, there also exist the negative AA and BB interactions which are much weaker than AB interaction. The magnetic behaviour of ferrimagnetic solids is explained on the basis of Neel's molecular field theory; the detailed description is beyond the scope of this book. Above the Curie temperature, the magnetization at both the sites obeys the Curie-Weiss law, whereas below this temperature the saturation effects come into existence and the magnetizations are obtained using Eq. (8.46).

# SOLVED EXAMPLES

**Example 8.1.** Calculate the diamagnetic susceptibility of atomic hydrogen in the ground state at S.T.P. using the wave function

$$\psi(r) = \frac{1}{\left(\pi a_0^3\right)^{1/2}} \exp\left(-\frac{r}{a_0}\right)$$

where $a_0$ (= 0.46Å) is the atomic radius.

**Solution.** The wave function for the ground state of hydrogen atom is

$$\psi(r) = \frac{1}{\left(\pi a_0^3\right)^{1/2}} \exp\left(-\frac{r}{a_0}\right)$$

The mean square distance of electronic charge distribution from the nucleus is calculated as

$$\langle r^2 \rangle = \int \Psi^* r^2 \Psi \, \mathbf{dr}$$

$$= 4\pi \int_0^\infty \psi^* r^2 \psi r^2 dr$$

$$= \frac{4\pi}{\pi a_0^3} \int_0^\infty r^4 \exp\left(-\frac{2r}{a_0}\right) dr$$

Put $$-\frac{2r}{a_0} = t$$

$\therefore$ $$r = -\frac{a_0}{2} t \quad \text{and} \quad dr = -\frac{a_0}{2} dt$$

$\therefore$ $$\langle r^2 \rangle = \frac{4}{a_0^3} \left(\frac{a_0^4}{16}\right)\left(-\frac{a_0}{2}\right) \int_0^\infty t^4 e^{-t} dt$$

Now

$$\int_0^\infty t^4 e^{-t} dt = 24$$

$\therefore$ $$\langle r^2 \rangle = 3a_0^2 \qquad \text{(in magnitude)}$$

From Eq. (8.28), we have

$$\chi_{dia} = -\frac{N\mu_0 Ze^2}{6m} \langle r^2 \rangle$$

$$= -\frac{N\mu_0 Ze^2}{2m} a_0^2$$

Here $$N = \frac{6.023 \times 10^{26}}{2.24 \times 10^{-2}} = 2.69 \times 10^{28} \text{ m}^{-3}$$

$$Z = 1$$

$$a_0 = 0.46 \times 10^{-10} \text{m}$$

$$\therefore \quad \chi_{dia} = -\frac{2.69 \times 10^{28} \times 4\pi \times 10^{-7} \times \left(1.6 \times 10^{-19}\right)^2 \left(0.46 \times 10^{-10}\right)^2}{2 \times 9.1 \times 10^{-31}}$$

$$= -1.01 \times 10^{-6} \text{ (SI units)}$$

**Example 8.2.** A paramagnetic gas with $L = 0$ and $S = 1/2$ contains $N$ atoms per $m^3$. Obtain expressions for the atomic densities in the possible energy levels and the resultant magnetization. If the gas contains $10^{28}$ atoms per $m^3$, calculate the populations of the levels and the magnetization at 300 K and 5 K for an applied magnetic flux density of 2.5 T.

**Solution.** Since $L = 0$, $S = 1/2$

$$\therefore \qquad J = 1/2,\ m_J = \pm \tfrac{1}{2} \text{ and } g = 2$$

Therefore, there are two energy levels with energies given by

$$E = m_J g \mu_B B$$

Let $N_{1/2}$ and $N_{-1/2}$ represent the populations of the levels with $m_J = 1/2$ and $-1/2$ respectively. Since population of a level $E$ is proportional to $\exp\left(-\frac{E}{kT}\right)$, we have

$$N_{1/2} = A \exp\left(-\frac{g\mu_B B}{2kT}\right); \quad N_{-1/2} = A \exp\left(\frac{g\mu_B B}{2kT}\right)$$

where A is a proportionality constant determined by the condition :

$$N_{1/2} + N_{-1/2} = N$$

It gives

$$A = \frac{N}{2\cosh\left(\frac{g\mu_B B}{2kT}\right)}; N_{1/2} = \frac{N\exp\left(-\frac{g\mu_B B}{2kT}\right)}{2\cosh\left(\frac{g\mu_B B}{2kT}\right)}; N_{-1/2} = \frac{N\exp\left(\frac{g\mu_B B}{2kT}\right)}{2\cosh\left(\frac{g\mu_B B}{2kT}\right)}$$

If $\mu$ is the atomic moment, the magnetization is given by

$$M = \mu\,(N_{-1/2} - N_{1/2}) = g\mu_B J\left(N_{-1/2} - N_{1/2}\right)$$

$$= g\mu_B JN\left[\frac{\exp\left(\frac{g\mu_B B}{2kT}\right) - \exp\left(-\frac{g\mu_B B}{2kT}\right)}{2\cosh\left(\frac{g\mu_B B}{2kT}\right)}\right]$$

$$= g\mu_B JN \tanh\left(\frac{g\mu_B B}{2kT}\right) \tag{8.83}$$

Now

$$g = 2,\ J = 1/2$$
$$\mu_B = 9.27 \times 10^{-24}\ \text{Am}^2$$
$$N = 10^{28}\ \text{m}^{-3}$$
$$B = 2.5\ \text{T}$$
$$k = 1.38 \times 10^{-23}\ \text{JK}^{-1}$$

The above expressions become

$$N_{1/2} = 5 \times 10^{27}\ \frac{\exp\left(\frac{1.68}{T}\right)}{\cosh\left(\frac{1.68}{T}\right)}$$

$$N_{-1/2} = 5 \times 10^{27}\ \frac{\exp\left(-\frac{1.68}{T}\right)}{\cosh\left(\frac{1.68}{T}\right)}$$

$$M = 9.27 \times 10^{4} \tanh\left(\frac{1.68}{T}\right)$$

At 300 K, we get

$$N_{1/2} = 5 \times 10^{27} \times \frac{0.994}{1.00} = 4.97 \times 10^{27}\ \text{m}^{-3}$$

$$N_{-1/2} = 5 \times 10^{27} \times \frac{1.006}{1.00} = 5.03 \times 10^{27}\ \text{m}^{-3}$$

$$M = 9.27 \times 10^{4} \times 5.6 \times 10^{-3} = 519.1\ \text{Am}^{-1}$$

Similarly, at 5 K, we obtain

$$N_{1/2} = 3.38 \times 10^{27}\ \text{m}^{-3}$$

$$N_{-1/2} = 6.62 \times 10^{27}\ \text{m}^{-3}$$

$$M = 3.0 \times 10^{4}\ \text{Am}^{-1}$$

**Example 8.3.** $Dy^{3+}$ has outer electronic configuration of $4f^9\ 6s^0$. Calculate the magnetic susceptibility for a salt containing one kg mole of $Dy^{3+}$ ions at 300 K.

**Solution.** Number of unpaired electrons in $Dy^{3+} = 5$.

Application of Hund's rules yields

$$S = 5(1/2) = 5/2$$

$$L = 3 + 2 + 1 + 0 - 1 = 5$$

$$J = L + S = 5/2 + 5 = 15/2$$

$$\therefore \qquad g = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)} = 1.33$$

Now

$$\chi = \frac{\mu_0 N \mu_B^2}{3kT} g^2 J(J+1)$$

For 1 kg mole of salt, $N = 6.023 \times 10^{26}$. Therefore,

$$\chi = \frac{4\pi \times 10^{-7} \times 6.023 \times 10^{26} \times \left(9.27 \times 10^{-24}\right)^2 \times (1.33)^2}{3 \times 1.38 \times 10^{-23} \times 300} \times \frac{15}{2}\left(\frac{15}{2} + 1\right)$$

$$= 5.9 \times 10^{-4}$$

**Example 8.4.** A ferromagnetic material with $J = 3/2$ and $g = 2$ has a Curie temperature of 125 K. Calculate the intrinsic flux density near 0 K. Also, calculate the ratio of the magnetization at 300 K in the presence of an external field of 1 mT to the spontaneous magnetization at 0 K.

**Solution.** At 0 K, the intrinsic flux density as given by Eqs. (8.52) and (8.59) is

$$B_E = \lambda M = \lambda M_s\,(0) = \lambda N g J \mu_B$$

Using Eq. (8.66), we get

$$B_E = \frac{3kT_C}{g\mu_B(J+1)} = \frac{3\times 1.38\times 10^{-23}\times 125}{2\times 9.27\times 10^{-24}\times(5/2)} = 111.6\ \text{T}$$

The magnetization for $T > T_C$ is obtained by using Eq. (8.64) as

$$M = \frac{CB}{\mu_o(T-T_C)}$$

Using Eqs. (8.65) and (8.66), we obtain

$$M = \frac{Ng^2\mu_B^2 J(J+1)B}{3k(T-T_C)} \tag{8.84}$$

Also,

$$M_s(0) = NgJ\mu_B$$

$$\therefore \quad \frac{M}{M_s(0)} = \frac{g\mu_B B(J+1)}{3k(T-T_C)}$$

$$= \frac{2\times 9.27\times 10^{-24}\times 10^{-3}\times(5/2)}{3\times 1.38\times 10^{-23}\times(300-125)}$$

$$= 6.4\times 10^{-6}$$

# SUMMARY

1. The magnetic moment per unit volume of a solid is called magnetization. The magnetization, **M**, produced per unit applied magnetic field, **H**, is called susceptibility, i.e.,

$$\chi = \mathbf{M/H}$$

2. The total magnetic flux density, **B**, inside a magnetic material is

$$\mathbf{B} = \mu_0\,(\mathbf{H} + \mathbf{M}) = \mu_0\,(1+\chi)\mathbf{H} = \mu_0\mu_r\,\mathbf{H} = \mu\mathbf{H}$$

where $\mu_0$, $\mu_r$ and $\mu$ are absolute permeability of free space, relative permeability of medium and absolute permeability of medium respectively.

3. Magnetism in solids arises mainly from the spins of unpaired electrons. The other minor contributions arise due to orbital motion of electrons and spins of nuclei.

4. Diamagnetic materials exhibit a small negative susceptibility given by

$$\chi_{dia} = -\frac{N\mu_0 Ze^2}{6m} \langle r^2 \rangle$$

where $N$ represents the number of atoms per unit volume of the solid, $Z$ the number of electrons present in each atom, $m$ the electronic mass and $\langle r^2 \rangle$ the mean square distance of electrons relative to the nucleus.

5. Paramagnetic materials possess small permanent magnetic moments which are oriented randomly in the absence of the field. The susceptibility is small but positive.

6. According to the classical theory, the paramagnetic susceptibility at normal field strengths and ordinary temperatures obeys the Curie law and is given by

$$\chi_{para} = \frac{\mu_0 N\mu^2}{3kT} = \frac{C}{T}$$

where $C \left( = \frac{\mu_0 N\mu^2}{3k} \right)$ is the Curie constant; $\mu$ being the magnetic moment of each atom and $k$ the Boltzmann's constant. The quantum theory gives an identical expression with $\mu^2$ replaced by $p^2_{eff}\,\mu^2_B$ where $\mu_B$ is the Bohr magneton and $p_{eff}$ the effective number of Bohr magnetons in the magnetic moment of a magnetic dipole. $p_{eff}$ is related to the total angular momentum quantum number $J$ of the dipole as

$$p_{eff} = g\sqrt{J(J+1)}$$

where $g$ is the Lande's $g$-factor.

6. Ferromagnetic materials consist of spontaneously magnetized regions called domains which are randomly oriented in the absence of an applied magnetic field. The spontaneous magnetization vanishes above a critical temperature called the Curie temperature when the ferromagnetic materials become paramagnetic.

7. Above the Curie temperature, $T_C$, the susceptibility of a ferromagnetic material obeys the Curie-Weiss law, i.e.,

$$\chi = \frac{C}{T - T_C}$$

where $C$ is the Curie-Weiss constant.

8. An antiferromagnetic material comprises two magnetic sublattices with equal and opposite magnetic moments such that there exists no net magnetization in the material. The magnetization, however, appears in the presence of an applied magnetic field. It increases with temperature and becomes maximum at a critical temperature called the Neel temperature, $T_N$. For $T > T_N$, the material behaves as paramagnetic.

9. A ferrimagnetic material is identical to an antiferromagnetic material except that the opposite magnetic moments of the neighbouring sublattices have unequal magnitudes. It also resembles the ferromagnetic material in the sense that both possess spontaneous magnetization and exhibit the phenomenon of hysteresis.

## VERY SHORT QUESTIONS

1. Define the following terms:

   (*i*) Magnetisation (*ii*) magnetic susceptibility (*iii*) magnetic flux density (*iv*) Permeability (*v*) Relative permeability.

2. Explain the meaning of the following terms in brief:

   (*i*) Diamagnetism (*ii*) Paramagnetism (*iii*) Ferromagnetism (*iv*) Ferrimagnetism (*v*) Antiferromagnetism.

3. Define the Curie law for paramgnetism.

4. What is the basic cause of paramagnetism?

5. What is spontaneous magnetisation?

6. What are domains?

7. What is exchange field?

8. Explain the following terms briefly :

   (*i*) Hysteresis (*ii*) Coercivity (*iii*) Remanence.

9. What is Neel temperature?

## SHORT QUESTIONS

1. What is diamagnetism? Why diamagnetic materials have negative susceptibility?

2. What is the essential difference between the classical theory and the quantum theory of paramagnetism?

3. Give the Curie law of paramagnetism. What is the Curie temperature?

4. Discuss the variation of spontaneous magnetisation with temperature for ferromagnetic materials.

5. What is the Curie-Weiss law? Discuss its application for ferromagnetic materials.
6. What is exchange interaction? How does it help to explain magnetism in iron group of atoms?
7. Explain the cause of hysteresis phenomenon in ferromagnetic materials? What does area of the loop signify?
8. Explain the difference between the terms 'Curie temperature' and 'Neel temperature'.
9. Explain the variation of susceptibility with temperature for antiferromagnetic materials.

## LONG QUESTIONS

1. Distinguish between the characteristic features of diamagnetism, paramagnetism, ferrimagnetism, antiferromagnetism and ferrimagnetism. Give an example of each type of material. Comment on the temperature variation of susceptibility for all types of materials.
2. Explain the origin of diamagnetism in materials. Obtain an expression for diamagnetic susceptibility using the Langevin's theory. What is the significance of negative susceptibility?
3. Derive an expression for diamagnetic susceptibility using the quantum theory. Discuss the temperature dependence of susceptibility.
4. Describe the Langevin's theory of paramagnetism and obtain an expression for paramagnetic susceptibility. Comment on the temperature dependence of susceptibility.
5. Give an account for the quantum theory of paramagnetism and discuss the low and high temperature cases. Explain how the orbital motion of electrons in transition and rare earth metals is quenched. Write the modified expression for susceptibility for the ions of iron group.
6. Give an account of the Weiss theory of ferromagnetism. Discuss the temperature variation of saturation magnetisation. Explain hysteresis and Curie point on the basis of this theory.
7. Explain how and why are the ferromagnetic domains formed? Draw a typical *B-H* loop and describe the different magnetisation processes, which lead to the formation of a *B-H* loop. What are the advantages and disadvantages of having a *B-H* loop in a material ?
8. Describe the Heisenberg's exchange interaction. How does it explain ferromagnetism? Relate the exchange integral to the Weiss constant and the ferromagnetic Curie temperature.

9. Describe the two-sublattice model to explain antiferromagnetism. How does this model account for the difference between the Neel temperature $T_N$ and the Curie-Weiss temperature, $\theta$?

## PROBLEMS

1. Calculate the diamagnetic susceptibility of copper assuming that an atom of copper contributes only one electron and the mean square distance of electronic distribution from the nucleus is equal to the radius of monovalent copper ion. The lattice constant of copper is 3.61 Å and the radius of $Cu^+$ is 0.96 Å. $(-4.62 \times 10^{-6})$

2. The electronic configuration of a $Cr^{2+}$ ion is $3d^4 4s^0$. Calculate the magnetic susceptibility for a salt containing one kg mole of $Cr^{2+}$ ions at 300 K. $(1.25 \times 10^{-4})$

   (**Hint** : Orbital angular momentum of $Cr^{2+}$ is quenched by the presence of the crystal field).

3. $Cu^{2+}$ has nine electrons in the $3d$-shell. What magnetic field must be applied to a salt containing $Cu^{2+}$ ions at 1 K so that 90 per cent of the ions are in the ground state ? (3 T)

4. The paramagnetism in copper sulphate arises mainly from the copper ions with $S = 1/2$. Show that the magnetization is given by the expression

$$M = N\mu_B \tanh\left(\frac{\mu_B B}{kT}\right)$$

5. Iron is a ferromagnet with the Curie temperature of 1043 K and an effective moment of 2.2 $\mu_B$ per atom. Compute the internal field for iron. (2100 T)

6. Europium oxide is ferromagnetic with the Curie temperature of 70 K. An europium ion has $J = 7/2$ and $g = 2$. Calculate the internal magnetic flux density. Also, determine the ratio of magnetization at 300 K in a field of 0.05 T to that at 0 K. $(34.7\text{ T}, 4.38 \times 10^{-4})$

7. Nickel (*fcc*) has a Curie temperature of 358°C and a lattice constant of 3.52 Å. A $Ni^{2+}$ ion has 8 electrons in the $3d$-shell. Assuming quenching of the orbital angular momentum and using the Hund's rules and the Weiss theory, determine the saturation magnetization, the Weiss field constant, the internal magnetic field and the Curie constant for nickel. $(1.7 \times 10^6\ \text{Am}^{-1}, 4.14 \times 10^{-4}, 704\text{ T}, 1.92)$